Sustainable Agriculture and Mitigating Shocks: A Panacea for Agricultural Productivity among Farming Households in Nigeria

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Abstract

Agriculture contributes about 23% of Nigeria's GDP, with over 70% of the population engaged in farming. Despite this involvement, most farming households remain poor, limiting productivity and making them vulnerable to shocks and natural hazards. This study assessed both farm and non-farm incomes generated by farmers in order to achieve sustainable agriculture as well as mitigate available shocks. The effect of farm and non-farm incomes in relation to the productivity of agricultural farming households in mitigating shocks was also examined. The Emergencies Monitoring Household Survey Year 2021 of the World Bank was used. Descriptive statistics were used to describe the socio-economic characteristics of farming households, the linear regression model was used in estimating the effect of farm income on agricultural productivity while the probit regression model was used in estimating the determinants of shocks. Gender ($p<0.1$), agricultural activity ($p<0.01$), residing in Zamfara State ($p<0.01$), residing in Katsina State ($p<0.05$), education

($p<0.05$), language ($p<0.01$) and age ($p<0.05$) were statistically significant predictors of productivity of farming households. While residing in Adamawa ($p<0.05$), Zamfara ($p<0.05$) and Katsina ($p<0.1$) states, age of household head ($p<0.01$), and engaging in both farm and non-farm activities ($p<0.01$) were predictors of shocks. On this premise, informed decisions supporting both farm and non-farm income generation should be critically weighed and supported by Nigerian agricultural sector stakeholders since farmers no longer depend solely on agriculturally-generated income to achieve sustainable agriculture and mitigate shocks.

Keywords: Agricultural productivity, mitigating shocks, farm income, non-farm income

JEL classification: D24, D31, E26, H27, H32, Q01

Introduction

Nigeria is the largest country in West Africa with a population of over 200 million people. The agricultural sector accounts for about 23% of Nigeria's gross domestic product (GDP)[1], with over 70% of the population involved in one agricultural activity or the other (Abiola & Adefabi, 2022; Akenbor & Esheya, 2022). Despite this huge involvement in the agricultural sector, the majority of farming households are still relatively poor (Berchoux et al., 2019). It is not easy therefore for these households to attain optimum agricultural productivity (Amare et al., 2018*)*. This also implies that the majority of farmers are easily prone to shocks and natural hazards assailing the agricultural sector.

Agricultural productivity is the rise in per capita output of agricultural produce within an economy over a period of time (Burodo, Babangida & Jobe, 2022), which could be measured monthly, quarterly or annually. Furthermore, agricultural productivity can also be the output produced by a given level of inputs in the agricultural sector of the economy over a given period of time (Awoyemi, Afolabi &Akomolafe, 2017). In a more technical term, this can be referred to as the value of total farm output to the value of total farm inputs (Burodo et al., 2022). Non-

[1] Agriculture in Nigeria — statistics and facts | Statista.

farm income earnings or proceeds generated from non-agricultural activities are a form of diversification from the income obtained from farming activities (Guest Editorial, 2024). In most cases, non-farm income serves as a form of boost or shock absorber to farming households. The ability to diversify or engage in other activities in addition to farming helps these households to reduce vulnerability to shocks and market fluctuations, thereby achieving sustainable agricultural practices (Mengistu and Belda, 2024). Some of the non-farm activities farming households engage in include trading (buying and selling), craftmanship, carpentry, commercial motorcycling, commercial bus driving, commercial gardening, artisanship and part-time labour (Pappas and Papadas, 2024). Besides the income generated from non-farm activities, remittances or income from migrant family members abroad can also be chanelled towards achieving successes from the agricultural sector by farming families.

In empirical studies, the primary theoretical framework explaining the linkage between nonfarm and farm sectors is articulated by Singh et al. (1986). They propose that households, considering their labour and resource endowments, make allocation decisions based on the sector offering the highest utility in returns. Consequently, households tend to allocate more labour to the sector with the most favourable returns. An important aspect of the relationship between the nonfarm sector and the farm sector is how household engagement in non-farm activities and the sale of their agricultural products impact this connection (Nkegbe et al., 2022). This study defines nonfarm activities as all endeavours undertaken by farming households away from their primary agricultural operations to generate additional income. Focusing exclusively on agricultural households, any economic pursuits they undertake outside of agriculture are categorized as nonfarm activities. These endeavours encompass a wide range of economic activities such as handicrafts, small-scale manufacturing (both household and non-household), construction, mining, quarrying, repair services, transportation, and petty trading.

When considering agricultural productivity, access to non-farm income, i.e., other sources of income besides farming, could affect the agricultural sector both positively and negatively (Sun, Li & Cui, 2024). Positively, it boosts access to inputs, reduces postharvest losses because farmers will be able to improvise or provide adequate storage for their farm produce thereby reducing losses (Mengistu and Belda, 2024). Non-farm income reduces dependence on agriculture, since the income generated can be used to augment purchase of inputs and equipment, expansion of soil and or improvement of soil fertility (Pappas and Papads, 2024). Having another source of income increases interactions with other people thereby improving social capita and people's net worth. On the negative side, farmers or farming households involved in non-farming activities in order to generate more income have the tendency to pay more attention to these activities than farming (Sun et al., 2024).

Problem Statement

Despite the importance of the agricultural sector to the Nigerian economy, the sector still faces many challenges (Abubakar and Attanda, 2013), critical amongst which are: use of outdated farming equipment and methods, inadequate infrastructure, limited access to credit and inputs, lack of access to information, and inconsistent government policies in relation to the agricultural sector (Korgbeelo, 2022). Modernization of the agricultural sector cannot be achieved without adequate funds (Fadeyi, 2021). Lack of credit and input as well as the need for farmers to mitigate certain shocks associated with farming activities, propels most farmers to engage in other money-making ventures besides farming (Anang, Nkrumah-Ennin & Nyaaba, 2020).

In 2021, over 8.7 million people in Adamawa, Borno and Yobe states needed humanitarian support as a result of farmer herder clashes, insecurity and the aftermath of the COVID-19 pandemic.[2] These states have been declared with heightened scenarios of food insecurity and hunger because the majority of occupants and inhabitants are mostly farmers. These states were the worse hit by insurgencies and farmer herder clashes. On the other hand, Katsina and Zamfara states were in the forefront of the struggle for natural resources control, banditry, farmer herder clashes and criminality. Because more than 70% of the inhabitants of these states are farmers, these inhumane actions have displaced more than 70,000 farmers from their original lands to neighbouring villages and

[2] North-east Nigeria: Borno, Adamawa and Yobe states Humanitarian Dashboard (January to March 2021) - Nigeria | ReliefWeb

communities and as far as Niger Republic, a neighbouring country.[3] This can in no wise enhance sustainable agricultural practices nor agricultural productivity in these states, hence the need for this research to assess how farming households in these five states have used non-farming income to mitigate shocks and achieve agricultural productivity.

Objectives of the study

The main objective of the study is to assess the effects of both farm and non-farm incomes on agricultural productivity and sustainable agriculture in order to mitigate shocks among farming households in Nigeria. The specific objectives are: to profile farming households and the non-farm incomes available to them, to estimate the effect of non-farm income on agricultural activities/productivity, and to identify shocks and ways farming households mitigate them.

Methodology

Data source

In order to adequately address the effects of non-farm income on agricultural productivity: Implication for sustainable agriculture and mitigating shocks among farming households in Nigeria, the data on Emergencies Monitoring Household Survey 2021 from the Food and Agriculture Organization (FAO), Data in Emergencies Hub, published on 8th February 2023 was used.[4] The FAO developed a monitoring system in 26 food crisis countries to better understand the impacts of various shocks on agricultural livelihoods, food security and local value chains. The Monitoring System consists of primary data collected from households on a periodic basis (more or less every four months, depending on seasonality). Between 20th May and 25th June 2021, FAO led a household survey of five states in Nigeria: Adamawa, Borno, Katsina, Yobe and Zamfara. These five states were considered based on their agrarian peculiarities and their susceptibility to shocks such as armed conflict and farmer herder clashes.[5] Adamawa, Borno and Yobe states are in the Northeastern region of Nigeria (Food and Agriculture Organization

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[3] Nigeria: Protection Monitoring Report - Katsina, Sokoto and Zamfara 3 - 15 January 2021 - Nigeria | ReliefWeb

[4] https://microdata.worldbank.org/index.php

[5] https://data-in-emergencies.fao.org/pages/monitoring

(FAO) in Nigeria, 2020), while Katsina and Zamfara are in the Northwestern region.

Sampling procedure

The sample design of this first round of data collection in Nigeria was Random Digital Dialing with no quota. The total sample size was 2739 households and the sample size per state was: 556 in Adamawa; 568 in Borno; 535 in Yobe, 758 in Katsina and 322 in Zamfara. Data were disaggregated for comparison into local government areas directly impacted by armed conflict and those not affected. The survey administered questionnaires covering the following information: household information, income sources, crop production, crop marketing, livestock production, livestock marketing, fisheries production, fisheries marketing, food security and consumption needs of households.

Analysis of data

To estimate its determinants, agricultural productivity was proxied as total income from farming households. This is the summation of both farm and non-farm incomes generated. In empirical studies, the primary theoretical framework explaining the linkage between nonfarm and farm sectors is articulated by Singh et al. (1986). Therefore, in order to estimate the relationship between agricultural productivity and determinant factors of agricultural productivity, the linear regression model was the best fit for estimation. The linear regression model useful for estimating linear relationships between variables gave a more accurate analysis.

Agricultural productivity (proxied as farm and non-farm) = total income (1)

$\sum(farm + nonfarm\ income)$ = function of explanatory variables (2)

i.e.

$$\sum Zi = f\,(X_1 + X_2 + X_3 + X_4 + \ldots\ldots\ldots\ldots X_n + e_n) \qquad (3)$$

the linear regression model,

where:

Zi = Agricultural Productivity (proxied as total income-sum total of farm and non-farm income)

X_1 = age of respondents

X_2 = gender (male =1, female = 0)

X_3 = resident type (resident = 1, non-resident/displaced =0)

X_4 = education (formal education =1, informal education =0)

X_5 = activity (non-agricultural and agricultural)

X_6 = language (Hausa =1, English = 0)

X_7 = Shocks (yes = 1, no = 0)

X_8 = prompt response after shock/disaster (yes =1, no = 0)

X_9 = Respondents' state (Borno=1, Yobe=2, Adamawa=3, Zamfara=4, Katsina=5)

In order to estimate the determinants of shocks among respondents in the study area, the probit regression model was used. Shocks were categorized as farmers affected (yes=1), or farmers not affected (no=0).

The probit regression model[6]

This is one of the regression types used in modelling dichotomous or binary outcome variables[7]. Binary outcome variables are those dependent variables with just two possibilities designated as yes or no (Nisbet, Elder & Miner, 2009; Nisbet, Miner & Yale, 2018).

To model a binary outcome, we have:

$$y^* = \alpha + \beta X + e \qquad (5)$$

$$Y_i = \begin{cases} 1 \; if \; y* > \; r \\ 0 \; if \; y* \leq \; r \end{cases} \qquad (6)$$

The Probit regression model is therefore specified as:

$$\Phi^{-1}(p_i) = \sum_{k=0}^{k=n} \beta KXik \qquad (7)$$

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[6] Probit Regression | Stata Data Analysis Examples (ucla.edu).

[7] Probit Model (Probit Regression): Definition - Statistics How To

Y = the dependent variable (Shocks: yes =1, no=0)

Ґ = the threshold;

B = vector of parameter

Φ = the normal distribution of errors

X = vector of explanatory variables

e_i = independent; distributed error term

X_1 = age (continuous variable)

X_2 = Gender (male =1, female= 0)

X_3 = Education (formal education =1, informal education =0)

X_4 = Total income (naira) (sum of farm and non-farm income)

X_5 = resident type (resident =1, non-resident/displaced=0)

X_6 = language (English =1, Hausa =0)

X_7 = activities (Agricultural and non-agricultural activities)

X_8 = were you called back after the shock/disaster (yes =1, no = 0)

X_9 = Respondents' state (Borno=1, Yobe=2, Adamawa=3, Zamfara=4, Katsina=5)

Variable specification

(i) Dependent variable: Exposure to Shocks. This study draws its inference from the studies of (Fujie & Senda, 2019; Tan et al., 2021). The study by Fujie and Senda was centred on an economic crisis considered as a man-made disaster which was characterized as an aggregate shock. The research focus is an investigation of the dynamics of productivity in pre-war rural Japan. In the case of Tan et al., their research was focused on the impact of climate change on rice yields in Malaysia. In their research, exposure to shocks (environmental — climate change, drought, floods — financial, health, and work-related) was classified into two major categories: exposure to shocks-yes; non-exposure to shocks-no;

(ii) Independent variables: All other variables and their indicators that could explain the exposure or non-exposure of farming households to shocks. These were age of respondents, residence status of respondents, language, involvement in farm and non-farm activities, gender of respondents.

Results and Discussion

Socio-economic characteristics of farming households

The socio-economic characteristics of the farming households are presented in Table 1. Eventually, we had 526 respondents in Adamawa (19.4%), 568 respondents in Borno (21%), 758 respondents in Katsina (28%), 532 in Yobe (19.7%) and 322 respondents in Zamfara (11.9%).

Table 1: Socio-economic characteristics of respondents

Variables	Frequency	Percent
State		
Adamawa	526	19.4
Borno	568	21.0
Katsina	758	28.0
Yobe	535	19.7
Zamfara	322	11.9
Gender		
Female	195	7.20
Male	2,514	92.80
Education		
Formal	2,443	90.18
No formal	266	9.82
Agricultural activities		
Yes - both crop and livestock production	2,184	80.62
No	525	19.38
Resident type		
Resident	2,527	93.28
Non-resident (displaced)	182	6.72
Language		
English	1520	43.89
Hausa	1189	56.11

Variables	Frequency	Percent
Response call during/after shock (call back by authorities)		
No	6	0.22
Yes	2703	99.78
Experienced shock/natural hazard		
Yes	1912	70.58
No	797	29.42

Source: https://microdata.fao.org

There were more males (92.80%) captured in the survey than females (7.20%). This is due to the cultural and religious background of the study area, where males are exposed to the public more than females. The majority of the respondents were fully agrarian (80.62%). At the time of this survey, they were plagued with shocks (70.58%) caused by natural disasters, farmer herder conflicts, droughts, etc. In order to fully understand the situation of farming households in the study area, we scrutinized their residence types. An overview of their type of residence showed that 93.28% were original occupants, i.e., residents of the areas surveyed, who had been living in the area longer than two years, after being displaced from their homes earlier. The remaining respondents were recent migrants (6.69%) who had been living in the present place of residence for less than two years. Almost all the households (99.78%) were called by the authorities during or after their displacement, while 70.58% experienced shocks or natural hazards.

The households' main sources of income were categorized into agricultural (crops and livestock production) and non-agricultural sources of income as shown in Figure 1. More than half (55.19%) of the farming households engage in non-farm activities as a way of augmenting income generated from farm activities, while 44.81% households obtain income from farming activities alone.

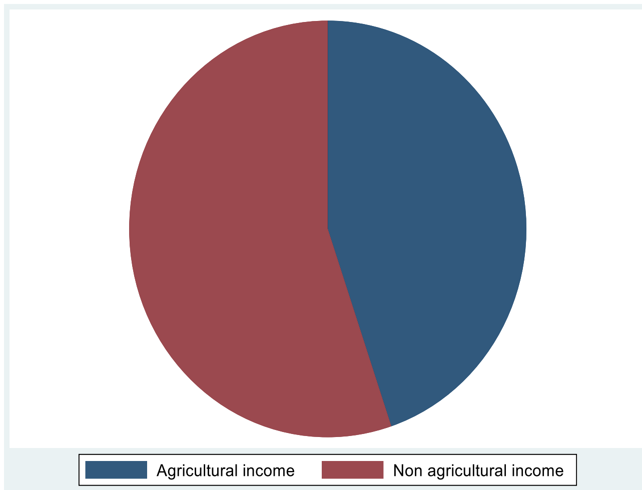

Figure 1: Source of income of respondents
(Agricultural Income-44.81%; Non-agricultural income-55.19%)
Source: https://microdata.fao.org

The socio-demographics of the households are captured in Table 2. The mean age of the farming households was 33.89±11.27 years, while the mean monthly total income was ₦277,076. The total income was a summation of all the incomes generated by the farming households. This included both agricultural and non-agricultural incomes (Table 2).

Table 2: Demographic characteristics of farming households

Variable	Observation	Mean	Standard deviation	Minimum	Maximum
Age	2709	33.89	11.27	17	89
Total income	2709	277,076	535455.9	0	8,7000,005

Source: https://microdata.fao.org

The analysis from the tables reveals that a significant portion of farming households experienced various shocks. While some households reported no shocks, a substantial proportion encountered multiple challenges. The shocks were mainly economic and health-related. High food and fuel prices were among the most prevalent economic shocks, affecting approximately 27.39% and 14.25% of households respectively (Table 3). These shocks strained household budgets and disrupted agricultural production processes. Sickness or death of the household head emerged as a notable concern, affecting 16.39% of the households. Such events can have profound socio-economic implications, impacting productivity and household stability. Pest outbreaks and animal diseases were identified as environmental challenges faced by farming communities, albeit with lower frequencies compared to economic and health-related shocks. Nevertheless, these events can have detrimental effects on crop yields and livestock health. The cumulative analysis underscores the compounding effects of multiple shocks on farming households. For instance, households affected by both high food prices and the sickness or death of the household head may face heightened vulnerabilities.

Table 3: Analysis of shocks faced by farming households

Shocks	Frequency	Percent
Exposure to Shocks		
No	1912	70.58
Yes	797	29.42
Sickness of Death of Household Head		
No	2265	83.61
Yes	444	16.39
Pest Outbreak		
No	2620	96.71
Yes	89	3.29
High Food Prices		
No	1967	72.61
Yes	742	27.39
High Fuel Prices		
No	2323	85.75
Yes	386	14.25
Can't Work or Do Business		
No	2502	92.36
Yes	207	7.64
Pest Outbreak		
No	2620	96.71
Yes	89	3.29

Shocks	Frequency	Percent
Animal Disease		
No	2617	96.60
Yes	92	3.40
Loss of Farm Employees		
No	2258	83.35
Yes	451	16.65
No Pasture		
No	2352	86.82
Yes	357	13.18
Other Economic Shocks		
No	2597	95.87
Yes	112	4.13

Source: https://microdata.fao.org

The effect of the environment on the activities of farming households in the study area is shown in Table 4. These environmental challenges faced by the farming households albeit with lower frequencies compared to economic and health-related shocks can become detrimental to the farmers, having an eventual negative effect if adequate attention is not paid to them. The environmental shocks most experienced by the farming households were due to the COVID-19 pandemic. These were: border closure (67.33%), problems with transportation of goods (50.61%), and market closure (44.63%). The other major challenge was violence and insecurity.

Table 4: Shocks experienced by farming households

Shocks	Frequencies	Percent
Drought		
No	2674	98.71
Yes	35	1.29
Flood		
No	2482	91.62
Yes	227	8.38
Fire		
No	2681	98.97
Yes	28	1.03

Shocks	Frequencies	Percent
Violence and Insecurity		
No	1861	68.70
Yes	848	31.30
Theft of Products or Assets		
No	2462	90.88
Yes	247	9.12
COVID-19 and Problem with Transport of goods		
No	1338	49.39
Yes	1371	50.61
Market Closure due to COVID-19		
No	1500	55.37
Yes	1209	44.63
Affected by Border Closure due to COVID-19		
No	885	32.67
Yes	1824	67.33

Source: https://microdata.fao.org

Determinants of Shocks

The determinants of shocks in the study area are presented in Table 5. The result of the probit regression reveals that the pseudo-R-squared is approximately 0.062, indicating that around 62% of the variance in exposure to shock is explained by the independent variables. Based on the statistics, the model appears to have a good explanatory power (pseudo-R-squared of 0.062) and a good fit to the data (significant chi-square and low p-value). Furthermore, from Table 6, it can be deduced that living in Adamawa (-0.198) and Zamfara (-0.218) and Kastina (-0.124) predisposes farming households to shocks. These variables are negative yet significant factors of exposure to shocks. It can also be deduced that older household heads are less likely to be prone to being exposed to shocks and other hazards than younger family members. This research outcome is similar to what was obtained in Brazil by Silva et al. (2017)) where younger farmers suffered more of the effects of drought, causing them to be more vulnerable. The language of respondents (0.386) was positive and significant at 1 percent. This shows that households that could speak both Hausa and English were more able to communicate effortlessly with the rescue team, international aid providers and extension agents than households whose mode of communication was only Hausa. Furthermore, the type of residence of households was also positive and significant (0.275), an indication that those who had been living in these states as permanent residents were able to respond to shocks and natural hazards quicker than those who were new to the states.

Table 5: Determinants of shocks

Probit regression

Exposure to Shock	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
: base Borno State	0	.	.	.	.	.	
Yobe	-0.075	0.082	-0.91	0.364	-0.236	0.086	
Adamawa	-0.196	0.082	-2.40	0.017	-0.357	-0.036	**
Zamfara	-0.218	0.094	-2.31	0.021	-0.403	-0.033	**
Katsina	-0.124	0.075	-1.65	0.099	-0.271	0.023	*
Age HH	-0.010	0.002	-4.25	0.000	-0.015	-0.006	***
Agric and non-farming activity HH	-0.376	0.064	-5.86	0.000	-0.501	-0.250	***
Education	0.097	0.094	1.03	0.301	-0.087	0.282	
Total income	0.000	0.000	1.29	0.196	0.000	0.000	
Callback	-0.205	0.532	-0.39	0.700	-1.249	0.838	
Language	0.386	0.056	6.95	0.000	0.277	0.496	***
Resident type HH	0.275	0.106	2.59	0.09	0.067	0.482	***
Constant	-0.446	0.591	-0.76	0.45	-1.604	0.711	

Source: https://microdata.fao.org; *** p<.01, ** p<.05, * p<.1

Determinants of Agricultural Productivity

The determinants of agricultural productivity (Table 6) were estimated using the ordinary least squares (OLS) method. The dependent variable was "Total-Income" with several independent variables tested for their effect on total income. The coefficients (Coef.) were estimated for each independent variable in the regression model. They represent the change in the dependent variable for a unit change in the independent variable, holding all other variables constant.

Table 6: Determinants of agricultural productivity

Linear regression

Total- Income	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
: base Borno State	0	.	.	.	.	.	
Yobe	-21788.598	32636.853	-0.67	0.504	-85784.385	42207.189	
Adamawa	-4505.525	32454.174	-0.14	0.890	-68143.107	59132.057	
Zamfara	102804.71	37207.991	2.76	0.006	29845.635	175763.79	***
Katsina	76682.75	29921.051	2.56	0.010	18012.227	135353.27	**
Agric-activity HH	110564.68	26409.982	4.19	0.000	58778.817	162350.54	***
Gender	68080.032	39852.159	1.71	0.088	-10063.847	146223.91	*
Education	87449.098	35371.052	2.47	0.013	18091.972	156806.23	**
Exposure to Shock	27245.869	22959.723	1.19	0.235	-17774.573	72266.312	
Callback	-56667.928	217616.45	-0.26	0.795	-483379.9	370044.04	
Language	59842.986	21802.893	2.74	0.006	17090.907	102595.07	***
Age	2043.555	924.3880	2.21	0.027	230.9750	3856.136	**
Resident type HH	33382.075	40791.535	0.82	0.413	-46603.774	113367.93	
Constant	-101489.32	239601.58	-0.42	0.672	-571310.71	368332.07	
Mean dependent var		277075.978	SD dependent var			535455.854	
R-squared		0.023	Number of observations			2709	
F-test		5.321	Prob > F			0.000	
Akaike crit. (AIC)		79117.549	Bayesian crit. (BIC)			79194.305	

Source: https://microdata.fao.org; *** p<0.01, ** p<0.05, * p<0.1

Table 6 indicates that gender ($p<0.1$), agricultural activity ($p<0.01$), residing in Zamfara State ($p<0.01$), residing in Katsina State ($p<0.05$), education ($p<0.05$), language ($p<0.01$) and age ($p<0.05$) were statistically significant predictors of total income, based on their coefficients and corresponding p-values, since their p-values are less than 0.05. Residing in Zamfara and Katsina states shows a positive relation with an increase in the total income of faming households compared to residing in Yobe and Adamawa states. This result is a reflection and a representation of the present situation in these states where economic activities are going on smoothly without any hindrance or fear (Afouda et al., 2019). Households that engaged in non-farm activities for additional income, in addition to agricultural activities, had a positive coefficient and higher total income. This is similar to the result obtained by Mazibuko and Antwi (2019), where additional income increased the productivity of farmers. This however differs from the research result of Sun et al. (2024), where non-farm income increased farmland abandonment and led to a reduction in agricultural production. The residence of respondents (state where farming households reside, Zamfara and Katsina) had a positive and significant relationship with increase in agricultural productivity. Households that were engaged in agricultural activity had a positive relationship with total income, the proxy of agricultural productivity. This implies that households that were solely involved in agricultural activities experienced increase in total income. The gender of households members had a positive effect on agricultural productivity. More males were involved in agricultural activities than females, a result of the culture of the region (Kamara, Kamai & Omoigui, 2020).

Conclusion and Recommendations

Overall, the combination of non-farm income opportunities, agricultural activities and enhanced agricultural productivity can mitigate shocks, reduce poverty, enhance food security, create employment, promote sustainable resource management, and foster economic development in these five states. The synergy between non-farm income and agricultural productivity is crucial for achieving sustainable development goals and improving the well-being of farming families in these states. It is important to also note that the potential of non-farm income to mitigate shocks and enhance agricultural productivity depends on various factors, including the availability of non-farm opportunities, the state of residence of farming households, gender, educational status, language, and whether respondents were called back after experiencing shock or natural disasters. Therefore, promoting non-farm income opportunities alongside agricultural development strategies will contribute to building absorbers to shocks, reduce vulnerability, and improve overall productivity among farmers.

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